

Power System Design Study

Christopher Gardner

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Section 1, Chapter 1: Power System Stability for Single Generator into an infinite Bus

Source: Energy Function Analysis For Power System Stability

Concept:

A power system includes generators, transmission lines (HVAC), loads (like home or factories), transformers, static var compensator (a rigid system that injects and absorbs power quickly like a mosfet), and HVDC lines. Usually, studies involve upwards of 10,000 nodes (buses) and a quarter will have generators. The main goal is to study contingency's—faults--and how to prevent them so that a entire system won't collapse from one fault. In order to analyze systems that remain stable with different faults/stresses a transient stability study is done. The transient stability studies the test if a system remains stable as time goes to infinity after a contingency/disturbance occurs.

Key Takeaways

- Generators, HVAC, HVDC, loads, transformers, and static var compensators make up a power grid
- Goal is to make sure a system remains operational after incidents
- Transient stability study is done to model a system under disturbance

PSSE Application

- Bus: Node where multiple branches occur
- Generator: Machine that ejects current into nearby buses
- Transmission Line: Branch with tap ratio or phase shift
- Load: MW/MVAR (Megawatts/Megavolt-amperes reactive) connected to bus
- SVC: static var compensator used to control voltage

Research Notes

- How do I know whether to use HVAC or HVDC
- What can I do to cause infinitesimal loss during transmission and step-down

Cont.

Concept:

Simulations can differ from operation due to contingencies and operating parameters. As such designers must do preventative measures called a DSA (dynamic security assessment). A DSA is a series of non-linear equation assessments, that run Transient stability or dynamic simulations that breaks down architecture to look for the most compromised parts. To circumvent the computing power of a DSA, engineers use TSI (Transient Stability Index) that can see the degree of stability a system has after disturbance numerically. To make it useful you can convert the findings into a normalized stability margin.

The math can be broken into two ways: one, time-simulation techniques for a set of contingencies and use the given output as guidelines, or two use the Lyapunov method to find the maximum energy a system can absorb before failure.

For DSA analysis both PEBS (Potential Energy Boundary Surface) and UEP (Unstable Equilibrium Point) methods are used to calculate margins and determine stability. Most DSA's use this in conjunction to Lyapunov's method

Key Takeaways

- DSA's are highly complex analysis methods requiring time and money but are very accurate and easy to read
- TSI's area able to find margins of operation with more linear mathematics
- Both of which are designed to study the event of a contingency

PSSE Application

- The academic license of PSSE can run a DSA with a limit of 50 buses
- The software requires python code to run the Lyapunov Method
- Its worthy to note both PEBS and UEP methods will be run by the DSA

Research Notes

- How can the Lyapunov method best judge the integration of PV+BSEE?
- What's the applicational difference between the two in my design study?

1.2: Types of Stability Problems

Concept:

There are two types of disturbances; small and big. Small disturbance can be analyzed with linear equations. However, big disturbance requires non-linear differential equations to solve, which can sometimes have algebraic constraints.

Stability is determined by after a disturbance if the system is able to return to a acceptable margin of operation equilibrium, which is the exact goal of transient stability calculations. As such when determining configuration or additions to a grid one takes in account protective measures can change how it is configured due to the vast differences between steady-state and disturbance. Therefore, to be considered transiently stable the system must retain synchronism (the synchronous generators continue rotating together at the same electrical frequency without losing their relative rotor angle relationship) for basic AC systems. Whilst DC systems may work even when loss of synchronism.

The definition therefore of Transient Stability is: the acceptable post-disturbance, steady-state operation of a grid as time increases.

Key Takeaways

- Two types of disturbance small and big
- Big is more complex, while small is algebra
- Stability is found if a system can return to normal after a event
- Retaining synchronism for AC is vital for Transient Stability
- Switching poles can sometimes still be okay for Transient Stability

PSSE Application

- Small Disturbance: Linearized dynamic studies, Modal analysis, small-signal stability
- Large Disturbance: Three-phase fault, generator trips, line outage, transient stability simulation
- Observed outputs: rotor angles, generator speeds, bus voltages, system frequencys

Research Notes

- My analysis of the grid post-integration should see after simulations that the grid (both AC and DC parts) can run stable
- What contingencies create the smallest and largest stability margins?
- How does PV inverters affect transient stability?

1.2.1: Synchronous Operations of Machines and Grid

Concept:

To see if a machine is synchronous with its environment of other machines or power networks the average rotor angular velocity and pole pairs must be equal to the angular frequency of the AC voltage delivered by the power network or of its environment.

A grid is synchronous if all connected machines are synched with the AC network and one another. The rotor angle is done with δ , relative angle must be measured to test for stability and should follow marginally the same logarithm path.

Key Takeaways

- Rotor angular velocity is how to tell if synchronized with angular frequency
- Little delta is used for relative angle

PSSE Application

- How can I exam machines in PSSE or PowerWorld for their post-disturbance activity.
- Rotor angle (δ)
- Rotor speed (ω)
- Electrical power output (P_e)
- Mechanical power (P_m)

Research Notes

- How can I exam machines to see how they would be affected post-disturbance?

Chapter 2: Transformers

Source: Power System Analysis

Concept:

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Key Takeaways

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PSSE Application

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Concept:

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Section 2, Chapter 1: Experiment Methodology

1.0: Executive Summary:

The primary focus of this design study is on the IEEE 39-Bus "New England" power system by replacing selected synchronous generators with photovoltaic (PV) generation and Battery Energy Storage Systems (BESS). The goal is to analyze how the system responds during both normal operating conditions and disturbances using transient stability simulations with WECC generic inverter models. I also plan to explore Direct Stability Assessment (DSA) methods, including the PEBS and UEPS techniques, if they can be implemented successfully within the scope of the project.

To complement the transmission study, OpenDSS and Python will be used to perform Quasi-Static Time Series (QSTS) simulations on a distribution feeder. These simulations will evaluate photovoltaic hosting capacity and investigate how Volt-VAR control and Static VAR Compensators (SVCs) can improve voltage regulation and overall system performance.

The primary focus of this study is transient stability analysis of the IEEE 39-Bus system following PV and BESS integration. Depending on project progress, complementary studies including N-1 contingency analysis, OPF, and OpenDSS-based hosting capacity may also be explored.

1.1: Research Question:

How does replacing synchronous generators with photovoltaic generators and battery energy storage system affect the transient stability of a representative IEEE 39-Bus Power System.

- How do rotor angles change after faults?
- How does frequency recovery compare?
- Which buses become voltage weak?
- How many generators can be replaced with PV+BESS before faults appear?
- What contingencies create the largest stability margin reduction?

1.2: Learning Objectives:

1. Develop skills with PSSE, PowerWorld, and OpenDSS.
2. Develop and validate the original IEEE 39-Bus transmission in PSSE.
3. Perform transient stability studies.
4. Evaluate system performance with metrics.

1.3: Scope:

- IEEE 39 Bus
- PSSE, PowerWorld, OpenDSS
- WECC Models
- Steady-state load flow
- Hosting capacity
- Contingency events
- Transient Stability

1.4: Methodology:

1. Build and validate IEEE 39-Bus system in PSSE
2. Perform baseline load-flow analysis
3. Run control load-flow analysis
4. Begin to replace selected generators with PV
5. Run load-flow analysis
6. Integrate BESS using WECC inverter models
7. Repeat simulations under identical disturbances
8. Compare before and after stability
9. Draw conclusions

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